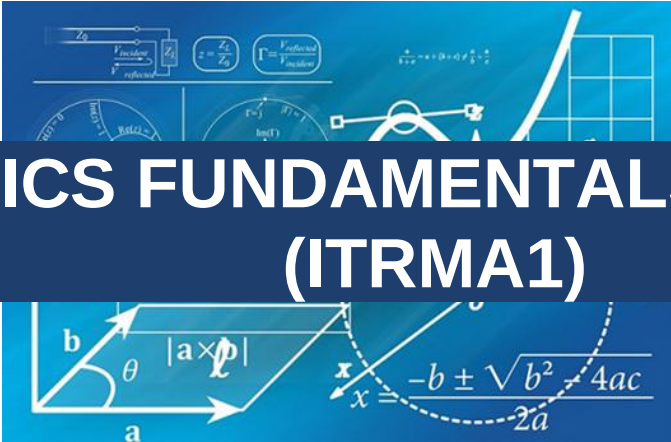




MATHEMATICS FUNDAMENTALS FOR ROBOTICS (ITRMA1)

MODULE NOTE 2025

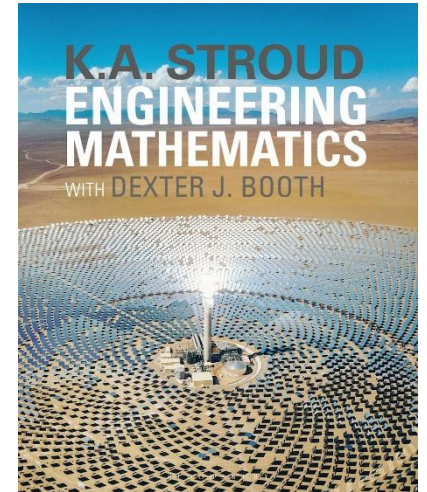
Module Code: **ITRMA1**

Session Number: **1-6**

Topic: **Solution of Differential Equations.**

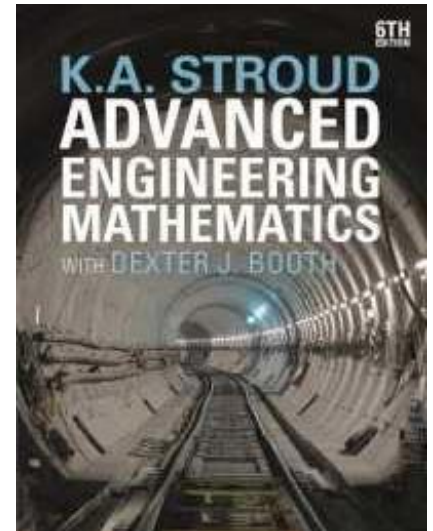
Prescribed reading:

Textbook: Stroud, K.A., 2020, Engineering Mathematics, 9781352010275, 8th edition, Bloomsbury Academic.



Recommended Reading:

Textbook: Stroud, K.A., 2003, Advanced Engineering Mathematics, 4th edition, Palgrave Macmillan.



Solution of differential equations

To solve a differential equation, we have to find the function for which the equation is true. This means that we have to manipulate the equation so as to eliminate all the derivatives and leave a relationship between y and x .

- **Method 1: By direct integration**

If the equation can be arranged in the form $\frac{dy}{dx} = f(x)$, then the equation can be solved by simple integration.

Example 1

$$\frac{dy}{dx} = 3x^2 - 6x + 5$$

Then $y = \int (3x^2 - 6x + 5)dx = x^3 - 3x^2 + 5x + C$

i.e. $y = x^3 - 3x^2 + 5x + C$

As always, of course, the constant of integration must be included. Here it provides the one arbitrary constant which we always get when solving a first-order differential equation.

Example 2

Solve $x \frac{dy}{dx} = 5x^3 + 4$

In this case, $\frac{dy}{dx} = 5x^2 + \frac{4}{x}$ So, $y = \dots\dots\dots$

$$y = \frac{5x^3}{3} + 4 \ln x + C$$

Example 3

Find the particular solution of the equation $e^x \frac{dy}{dx} = 4$, given that $y = 3$ when $x = 0$.

First rewrite the equation in the form $\frac{dy}{dx} = \frac{4}{e^x} = 4e^{-x}$.

Then $y = \int 4e^{-x} dx = -4e^{-x} + C$

Knowing that when $x = 0$, $y = 3$, we can evaluate C in this case, so that the required particular solution is $y = \dots\dots\dots$

$$y = -4e^{-x} + 7$$

- **Method 2: By separating the variables.**

If the given equation is of the form $\frac{dy}{dx} = f(x, y)$, the variable y on the right-hand side prevents solving by direct integration. We therefore have to devise some other method of solution.

Let us consider equations of the form $\frac{dy}{dx} = f(x).F(y)$ and of the form $\frac{dy}{dx} = \frac{f(x)}{F(y)}$, i.e. equations in which the right-hand side can be expressed as products or quotients of functions of x or of y .

A few examples will show how we proceed.

Example 1

Solve $\frac{dy}{dx} = \frac{2x}{y+1}$

We can rewrite this as $(y+1) \frac{dy}{dx} = 2x$

Now integrate both sides with respect to x :

$$\int (y+1) \frac{dy}{dx} dx = \int 2x dx \quad \text{i.e.}$$

and this gives $\frac{y^2}{2} + y = x^2 + C$

$$\int (y + 1) dy = \int 2x dx$$

Example 2

Solve $\frac{dy}{dx} = (1 + x)(1 + y)$

$$\frac{1}{1 + y} \frac{dy}{dx} = 1 + x$$

Integrate both sides with respect to x :

$$\int \frac{1}{1 + y} \frac{dy}{dx} dx = \int (1 + x) dx \quad \therefore \int \frac{1}{1 + y} dy = \int (1 + x) dx$$

$$\ln(1 + y) = x + \frac{x^2}{2} + C$$

The method depends on our being able to express the given equation in the form $F(y) \cdot \frac{dy}{dx} = f(x)$. If this can be done, the rest is then easy, for we have

$$\int F(y) \cdot \frac{dy}{dx} dx = \int f(x) dx \quad \therefore \int F(y) dy = \int f(x) dx$$

and we then continue as in the examples.

Example 3

Solve $\frac{dy}{dx} = \frac{1+y}{2+x}$ (1)

This can be written as $\frac{1}{1+y} \frac{dy}{dx} = \frac{1}{2+x}$

Integrate both sides with respect to x :

$$\begin{aligned}\int \frac{1}{1+y} \frac{dy}{dx} dx &= \int \frac{1}{2+x} dx \\ \therefore \int \frac{1}{1+y} dy &= \int \frac{1}{2+x} dx \\ \therefore \ln(1+y) &= \ln(2+x) + C\end{aligned}\quad (2)$$

It is convenient to write the constant C as the logarithm of some other constant A :

$$\begin{aligned}\ln(1+y) &= \ln(2+x) + \ln A = \ln A(2+x) \\ \therefore 1+y &= A(2+x)\end{aligned}$$

Note: We can, in practice, get from the given equation (1) to the form of the equation in (2) by a simple routine, thus:

$$\frac{dy}{dx} = \frac{1+y}{2+x}$$

First multiply across by dx:

$$dy = \frac{1+y}{2+x} dx$$

Now collect the 'y-factor' with the dy on the left, i.e. divide by (1 + y):

$$\frac{1}{1+y} dy = \frac{1}{2+x} dx$$

Finally, add the integral signs:

$$\int \frac{1}{1+y} dy = \int \frac{1}{2+x} dx$$

and then continue as before.

This is purely a routine which enables us to sort out the equation algebraically, the whole of the work being done in one line. Notice, however, that the RHS of the given equation must be expressed as 'x-factors' and 'y-factors'.

Example 4

Solve $\frac{dy}{dx} = \frac{y^2 + xy^2}{x^2y - x^2}$

First express the RHS in 'x-factors' and 'y-factors':

$$\frac{dy}{dx} = \frac{y^2(1+x)}{x^2(y-1)}$$

Now rearrange the equation so that we have the 'y-factors' and dy on the LHS and the 'x-factors' and dx on the RHS.

$$\frac{y-1}{y^2} dy = \frac{1+x}{x^2} dx$$

We now add the integral signs:

$$\int \frac{y-1}{y^2} dy = \int \frac{1+x}{x^2} dx$$

and complete the solution:

$$\int \left\{ \frac{1}{y} - y^{-2} \right\} dy = \int \left\{ x^{-2} + \frac{1}{x} \right\} dx$$

$$\therefore \ln y + y^{-1} = \ln x - x^{-1} + C$$

$$\therefore \ln y + \frac{1}{y} = \ln x - \frac{1}{x} + C$$

Example 5

Solve

$$\frac{dy}{dx} = \frac{y^2 - 1}{x}$$

$$dy = \frac{y^2 - 1}{x} dx$$

Rearranging, we have

$$\frac{1}{y^2 - 1} dy = \frac{1}{x} dx$$

$$\int \frac{1}{y^2 - 1} dy = \int \frac{1}{x} dx$$

$$\frac{1}{2} \ln \frac{y-1}{y+1} = \ln x + C$$

$$\therefore \ln \frac{y-1}{y+1} = 2 \ln x + \ln A$$

$$\therefore \frac{y-1}{y+1} = Ax^2$$

$$y - 1 = Ax^2(y + 1)$$

Exercises

Find the general solutions of the following equations:

$$1 \quad \frac{dy}{dx} = \frac{y}{x}$$

$$2 \quad \frac{dy}{dx} = (y + 2)(x + 1)$$

$$3 \quad \cos^2 x \frac{dy}{dx} = y + 3$$

$$4 \quad \frac{dy}{dx} = xy - y$$

$$5 \quad \frac{\sin x}{1 + y} \cdot \frac{dy}{dx} = \cos x$$

Answers to the exercises

$$\begin{aligned} 1 \quad \frac{dy}{dx} &= \frac{y}{x} \quad \therefore \int \frac{1}{y} dy = \int \frac{1}{x} dx \\ \therefore \ln y &= \ln x + C \\ &= \ln x + \ln A \\ \therefore y &= Ax \end{aligned}$$

$$\begin{aligned} 2 \quad \frac{dy}{dx} &= (y+2)(x+1) \\ \therefore \int \frac{1}{y+2} dy &= \int (x+1) dx \\ \therefore \ln(y+2) &= \frac{x^2}{2} + x + C \end{aligned}$$

$$\begin{aligned} 3 \quad \cos^2 x \frac{dy}{dx} &= y+3 \\ \therefore \int \frac{1}{y+3} dy &= \int \frac{1}{\cos^2 x} dx \end{aligned}$$

$$= \int \sec^2 x \, dx$$

$$\ln(y + 3) = \tan x + C$$

$$4 \quad \frac{dy}{dx} = xy - y \quad \therefore \frac{dy}{dx} = y(x - 1)$$

$$\therefore \int \frac{1}{y} dy = \int (x - 1) dx$$

$$\therefore \ln y = \frac{x^2}{2} - x + C$$

$$5 \quad \frac{\sin x}{1 + y} \cdot \frac{dy}{dx} = \cos x$$

$$\int \frac{1}{1 + y} dy = \int \frac{\cos x}{\sin x} dx$$

$$\therefore \ln(1 + y) = \ln \sin x + C$$

$$= \ln \sin x + \ln A$$

$$1 + y = A \sin x$$

$$\therefore y = A \sin x - 1$$

Method 3: Homogeneous equations – by substituting $y = vx$.

Here is an equation:

$$\frac{dy}{dx} = \frac{x + 3y}{2x}$$

This looks simple enough, but we find that we cannot express the RHS in the form of 'x-factors' and 'y-factors', so we cannot solve by the method of separating the variables.

In this case we make the substitution $y = vx$, where v is a function of x . So $y = vx$. Differentiate with respect to x (using the product rule):

$$\therefore \frac{dy}{dx} = v \cdot 1 + x \frac{dv}{dx} = v + x \frac{dv}{dx}$$

$$\text{Also } \frac{x + 3y}{2x} = \frac{x + 3vx}{2x} = \frac{1 + 3v}{2}$$

$$\text{The equation now becomes } v + x \frac{dv}{dx} = \frac{1 + 3v}{2}$$

$$\begin{aligned} \therefore x \frac{dv}{dx} &= \frac{1 + 3v}{2} - v \\ &= \frac{1 + 3v - 2v}{2} = \frac{1 + v}{2} \end{aligned}$$

$$\therefore x \frac{dv}{dx} = \frac{1 + v}{2}$$

The given equation is now expressed in terms of v and x , and in this form we find that we can solve by separating the variables. Here goes:

$$\int \frac{2}{1+v} dv = \int \frac{1}{x} dx$$

$$\therefore 2\ln(1+v) = \ln x + C = \ln x + \ln A$$

$$(1+v)^2 = Ax$$

$$\text{But } y = vx \quad \therefore v = \left\{ \frac{y}{x} \right\} \quad \therefore \left(1 + \frac{y}{x} \right)^2 = Ax$$

$$\text{which gives } (x+y)^2 = Ax^3$$

Note: $\frac{dy}{dx} = \frac{x+3y}{2x}$ is an example of a *homogeneous differential equation*.

This is determined by the fact that the total degree in x and y for each of the terms involved is the same (in this case, of degree 1). The key to solving every homogeneous equation is to substitute $y = vx$ where v is a function of x . This converts the equation into a form which we can solve by separating the variables.

Example 1

Solve $\frac{dy}{dx} = \frac{x^2 + y^2}{xy}$

Here, all terms of the RHS are of degree 2, i.e. the equation is homogeneous. $\therefore$ We substitute $y = vx$ (where v is a function of x)

$$\therefore \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\text{and } \frac{x^2 + y^2}{xy} = \frac{x^2 + v^2 x^2}{vx^2} = \frac{1 + v^2}{v}$$

The equation now becomes:

$$v + x \frac{dv}{dx} = \frac{1 + v^2}{v}$$

$$\therefore x \frac{dv}{dx} = \frac{1 + v^2}{v} - v$$

$$= \frac{1 + v^2 - v^2}{v} = \frac{1}{v}$$

$$\therefore x \frac{dv}{dx} = \frac{1}{v}$$

Now you can separate the variables and get the result in terms of v and x .

$$\int v \, dv = \int \frac{1}{x} \, dx$$

$$\therefore \frac{v^2}{2} = \ln x + C$$

All that remains now is to express v back in terms of x and y . The substitution we used was $y = vx$ $\therefore v = \frac{y}{x}$

$$\therefore \frac{1}{2} \left(\frac{y}{x} \right)^2 = \ln x + C$$

$$y^2 = 2x^2(\ln x + C)$$

Example 2

Solve $\frac{dy}{dx} = \frac{2xy + 3y^2}{x^2 + 2xy}$

Is this a homogeneous equation? If you think so, what are your reasons?

Yes, because the degree of each term is the same

They are all, of course, of degree 2.

So we now make the substitution, $y = vx$, where v is a function of x

$$\begin{aligned} \frac{dy}{dx} &= \dots\dots\dots v + x \frac{dv}{dx} \\ \text{and } \frac{2xy + 3y^2}{x^2 + 2xy} &= \dots\dots\dots \frac{2vx^2 + 3v^2x^2}{x^2 + 2vx^2} = \frac{2v + 3v^2}{1 + 2v} \end{aligned}$$

So that $v + x \frac{dv}{dx} = \frac{2v + 3v^2}{1 + 2v}$

Now take the single v over to the RHS and simplify, giving:

$$\begin{aligned}
 x \frac{dv}{dx} &= \frac{2v + 3v^2}{1 + 2v} - v \\
 &= \frac{2v + 3v^2 - v - 2v^2}{1 + 2v}
 \end{aligned}$$

$$x \frac{dv}{dx} = \frac{v + v^2}{1 + 2v}$$

Now you can separate the variables, giving $\int \frac{1 + 2v}{v + v^2} dv = \int \frac{1}{x} dx$

$$\ln(v + v^2) = \ln x + C = \ln x + \ln A$$

$$\therefore v + v^2 = Ax$$

Remember the substitution was $y = vx$, so that $v = \frac{y}{x}$

$$v + v^2 = Ax \text{ and } v = \frac{y}{x} \quad \therefore \frac{y}{x} + \frac{y^2}{x^2} = Ax$$

$$xy + y^2 = Ax^3$$

Example 3

$$\text{Solve } (x^2 + y^2) \frac{dy}{dx} = xy$$

$$(x^2 + y^2) \frac{dy}{dx} = xy \quad \therefore \frac{dy}{dx} = \frac{xy}{x^2 + y^2}$$

$$\text{Put } y = vx \quad \therefore \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\text{and } \frac{xy}{x^2 + y^2} = \frac{vx^2}{x^2 + v^2x^2} = \frac{v}{1 + v^2}$$

$$\therefore v + x \frac{dv}{dx} = \frac{v}{1 + v^2}$$

$$x \frac{dv}{dx} = \frac{v}{1 + v^2} - v$$

$$x \frac{dv}{dx} = \frac{v - v - v^3}{1 + v^2} = \frac{-v^3}{1 + v^2}$$

$$\therefore \int \frac{1 + v^2}{v^3} dv = - \int \frac{1}{x} dx$$

$$\therefore \int \left(v^{-3} + \frac{1}{v} \right) dv = -\ln x + C$$

$$\therefore \frac{-v^{-2}}{2} + \ln v = -\ln x + \ln A$$

$$\ln v + \ln x + \ln K = \frac{1}{2v^2} \quad (\ln K = -\ln A)$$

$$\ln Kvx = \frac{1}{2v^2}$$

$$\text{But } v = \frac{y}{x} \quad \therefore \ln Ky = \frac{x^2}{2y^2}$$

$$2y^2 \ln Ky = x^2$$

Exercises

Solve the following:

1 $(x - y) \frac{dy}{dx} = x + y$

2 $2x^2 \frac{dy}{dx} = x^2 + y^2$

3 $(x^2 + xy) \frac{dy}{dx} = xy - y^2$

Answers to the exercises

$$1 \quad (x - y) \frac{dy}{dx} = x + y \quad \therefore \frac{dy}{dx} = \frac{x + y}{x - y}$$

$$\text{Put } y = vx \quad \therefore \frac{dy}{dx} = v + x \frac{dv}{dx} \quad \frac{x + y}{x - y} = \frac{1 + v}{1 - v}$$

$$\therefore v + x \frac{dv}{dx} = \frac{1 + v}{1 - v} \quad \therefore x \frac{dv}{dx} = \frac{1 + v}{1 - v} - v = \frac{1 + v - v + v^2}{1 - v} = \frac{1 + v^2}{1 - v}$$

$$\therefore \int \frac{1 - v}{1 + v^2} dv = \int \frac{1}{x} dx \quad \therefore \int \left\{ \frac{1}{1 + v^2} - \frac{v}{1 + v^2} \right\} dv = \ln x + C$$

$$\therefore \tan^{-1} v - \frac{1}{2} \ln(1 + v^2) = \ln x + \ln A$$

$$\text{But } v = \frac{y}{x} \quad \therefore \tan^{-1} \left\{ \frac{y}{x} \right\} = \ln A + \ln x + \frac{1}{2} \ln \left(1 + \frac{y^2}{x^2} \right)$$

$$2 \quad 2x^2 \frac{dy}{dx} = x^2 + y^2 \quad \therefore \frac{dy}{dx} = \frac{x^2 + y^2}{2x^2}$$

$$\text{Put } y = vx \quad \therefore \frac{dy}{dx} = v + x \frac{dv}{dx}; \quad \frac{x^2 + y^2}{2x^2} = \frac{x^2 + v^2 x^2}{2x^2} = \frac{1 + v^2}{2}$$

$$\therefore v + x \frac{dv}{dx} = \frac{1 + v^2}{2} \quad \therefore x \frac{dv}{dx} = \frac{1 + v^2}{2} - v = \frac{1 - 2v + v^2}{2} = \frac{(v - 1)^2}{2}$$

$$\therefore \int \frac{2}{(v - 1)^2} dv = \int \frac{1}{x} dx \quad \therefore -2 \frac{1}{v - 1} = \ln x + C$$

$$\text{But } v = \frac{y}{x} \text{ and } \frac{2}{1 - v} = \ln x + C \quad \therefore \frac{2x}{x - y} = \ln x + C$$

$$3 \quad (x^2 + xy) \frac{dy}{dx} = xy - y^2 \quad \therefore \frac{dy}{dx} = \frac{xy - y^2}{x^2 + xy}$$

$$\text{Put } y = vx \quad \therefore \frac{dy}{dx} = v + x \frac{dv}{dx}; \quad \frac{xy - y^2}{x^2 + xy} = \frac{vx^2 - v^2x^2}{x^2 + vx^2} = \frac{v - v^2}{1 + v}$$

$$\therefore v + x \frac{dv}{dx} = \frac{v - v^2}{1 + v}$$

$$x \frac{dv}{dx} = \frac{v - v^2}{1 + v} - v = \frac{v - v^2 - v - v^2}{1 + v} = \frac{-2v^2}{1 + v}$$

$$\therefore \int \frac{1 + v}{v^2} dv = \int \frac{-2}{x} dx$$

$$\int \left(v^{-2} + \frac{1}{v} \right) dv = - \int \frac{2}{x} dx$$

$$\therefore \ln v - \frac{1}{v} = -2 \ln x + C \quad \text{Let } C = \ln A$$

$$\ln v + 2 \ln x = \ln A + \frac{1}{v}$$

$$\ln \left\{ \frac{y}{x} \cdot x^2 \right\} = \ln A + \frac{x}{y} \quad \therefore xy = Ae^{x/y}$$

1. Intro to Differential Equations (DEs)

Method 4: Linear equations – use of integrating factor.

Consider the equation $\frac{dy}{dx} + 5y = e^{2x}$

This is clearly an equation of the first order, but different from those we have dealt with so far. In fact, none of our previous methods could be used to solve this one, so we have to find a further method of attack.

In this case, we begin by multiplying both sides by e^{5x} . This gives

$$e^{5x} \frac{dy}{dx} + y5e^{5x} = e^{2x} \cdot e^{5x} = e^{7x}$$

We now find that the LHS is, in fact, the derivative of $y \cdot e^{5x}$.

$$\therefore \frac{d}{dx} \left\{ y \cdot e^{5x} \right\} = e^{7x}$$

Now, of course, the rest is easy. Integrate both sides with respect to x :

$$\therefore y \cdot e^{5x} = \int e^{7x} dx = \frac{e^{7x}}{7} + C \quad \therefore y = \dots\dots\dots$$

$$y = \frac{e^{2x}}{7} + Ce^{-5x}$$

The equation we have just solved is an example of a set of equations of the form $\frac{dy}{dx} + Py = Q$, where P and Q are functions of x (or constants). This equation is called a *linear equation of the first order* and to solve any such equation, we multiply both sides by an *integrating factor* which is always $e^{\int P dx}$. This converts the LHS into the derivative of a product.

Consider $\frac{dy}{dx} + Py = Q$ where P and Q are functions of x . Integrating factor, $IF = e^{\int P dx}$

$$\therefore \frac{dy}{dx} \cdot e^{\int P dx} + Pye^{\int P dx} = Qe^{\int P dx}$$

You will now see that the LHS is the derivative of $ye^{\int P dx}$

$$\therefore \frac{d}{dx} \left\{ ye^{\int P dx} \right\} = Qe^{\int P dx}$$

Integrate both sides with respect to x :

$$ye^{\int P dx} = \int Qe^{\int P dx} \cdot dx$$

This result looks far more complicated than it really is. If we indicate the integrating factor by IF, this result becomes:

$$y \cdot IF = \int Q \cdot IF \, dx$$

So, the solution of an equation of the form

$$\frac{dy}{dx} + Py = Q \text{ (where } P \text{ and } Q \text{ are functions of } x\text{)}$$

$$\text{is given by } y \cdot IF = \int Q \cdot IF \, dx \quad \text{where } IF = e^{\int P dx}$$

Example 1

To solve $\frac{dy}{dx} - y = x$

If we compare this with $\frac{dy}{dx} + Py = Q$, we see that in this case
 $P = -1$ and $Q = x$.

The integrating factor is always $e^{\int P dx}$ and here $P = -1$.

$\therefore \int P dx = -x$ and the integrating factor is therefore

$$\boxed{e^{-x}}$$

We therefore multiply both sides by e^{-x} .

$$\therefore e^{-x} \frac{dy}{dx} - ye^{-x} = xe^{-x}$$

$$\frac{d}{dx} \left\{ e^{-x} y \right\} = xe^{-x} \quad \therefore ye^{-x} = \int xe^{-x} dx$$

The RHS integral can now be determined by integrating by parts:

$$ye^{-x} = x(-e^{-x}) + \int e^{-x} dx = -xe^{-x} - e^{-x} + C$$

$$\therefore y = -x - 1 + Ce^x$$

$$\therefore y = Ce^x - x - 1$$

Example 2

Solve $x \frac{dy}{dx} + y = x^3$

First we divide through by x to reduce the first term to a single $\frac{dy}{dx}$

i.e. $\frac{dy}{dx} + \frac{1}{x} \cdot y = x^2$

Compare with $\left[\frac{dy}{dx} + Py = Q \right]$

$\therefore P = \frac{1}{x}$ and $Q = x^2$

IF = $e^{\int P dx} = \int \frac{1}{x} dx = \ln x$

$\therefore$ IF = $e^{\ln x} = x$

$\therefore$ IF = x

The solution is $y \cdot \text{IF} = \int Q \cdot \text{IF} dx$ so $yx = \int x^2 \cdot x dx = \int x^3 dx = \frac{x^4}{4} + C$

$\therefore xy = \frac{x^4}{4} + C$

Example 3

Solve $\frac{dy}{dx} + y \cot x = \cos x$

Compare with $\left[\frac{dy}{dx} + Py = Q \right] \quad \therefore \begin{cases} P = \cot x \\ Q = \cos x \end{cases}$

$$\text{IF} = e^{\int P dx} \quad \int P dx = \int \cot x dx = \int \frac{\cos x}{\sin x} dx = \ln \sin x$$

$$\therefore \text{IF} = e^{\ln \sin x} = \sin x$$

$$y \cdot \text{IF} = \int Q \cdot \text{IF} dx \quad \therefore y \sin x = \int \sin x \cos x dx = \frac{\sin^2 x}{2} + C$$

$$\therefore y = \frac{\sin x}{2} + C x$$

Example 4

Solve $(x + 1) \frac{dy}{dx} + y = (x + 1)^2$

Divide through by $(x + 1)$

$$\therefore \frac{dy}{dx} + \frac{1}{x + 1} \cdot y = x + 1$$

Compare with $\frac{dy}{dx} + Py = Q$

In this case $P = \frac{1}{x + 1}$ and $Q = x + 1$

Now determine the integrating factor

$$\int P \, dx = \int \frac{1}{x + 1} \, dx = \ln(x + 1)$$

$$\therefore \text{IF} = e^{\ln(x+1)} = (x + 1)$$

The solution is always $y \cdot \text{IF} = \int Q \cdot \text{IF} \, dx$

and we know that, in this case, $\text{IF} = x + 1$ and $Q = x + 1$.

$$y \cdot (x + 1) = \int (x + 1)(x + 1) \, dx$$

$$= \int (x + 1)^2 \, dx$$

$$= \frac{(x + 1)^3}{3} + C$$

$$\therefore y = \frac{(x + 1)^2}{3} + \frac{C}{x + 1}$$

Exercises

Solve the following:

$$1 \quad \frac{dy}{dx} + 3y = e^{4x}$$

$$2 \quad x \frac{dy}{dx} + y = x \sin x$$

$$3 \quad \tan x \frac{dy}{dx} + y = \sec x$$

Answers to the exercises

$$1 \quad y = \frac{e^{4x}}{7} + Ce^{-3x} \quad (\text{IF} = e^{3x})$$

$$2 \quad xy = \sin x - x \cos x + C \quad (\text{IF} = x)$$

$$3 \quad y \sin x = x + C \quad (\text{IF} = \sin x)$$

- **Questions and Answers**



References

- [1] Stroud, K.A., 2020, Engineering Mathematics, 9781352010275, 8th edition, Bloomsbury Academic.
- [2] Stroud, K.A., 2003, Advanced Engineering Mathematics, 4th edition, Palgrave Macmillan.